ANSWER KEY & MARKING SCHEME SET A

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** physical change	no new substance forms
2	**(a)** chemical change	iron oxide forms
3	**(a)** combustion	burning
4	**(c)** a chemical change in their body	bioluminescence
5	**(d)** Freezing of water	ice melts back
6	**(b)** crystallisation	physical change
7	**(d)** Weathering of rocks	takes years
8	**(c)** desirable	useful
9	**(c)** physical, chemical	both occur
10	**(a)** cutting off oxygen supply	stops combustion

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Physical: no new substance (e.g. ice melting) (1); chemical: new substance forms (e.g. burning paper) (1).	1+1
12	Reaction with oxygen giving heat/light (1); wood and LPG (1).	1+1
13	The lowest temperature at which a substance catches fire (2).	2
14	Light produced by living organisms without heat (1); fireflies (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Burn the ribbon with tongs (1); white powder forms with a bright light (1); new substance magnesium oxide (1).	1+1+1
16	Fuel (1); oxygen (1); ignition temperature (1).	1+1+1
17	Physical: melting butter, dissolving salt (1.5); chemical: burning candle, cooking egg (1.5).	1½+1½
	OR: Pass gas through lime water (1); turns milky (1); confirms CO_2 (1).	1+1+1
18	Weakens iron objects (1.5); painting, oiling, galvanising (1.5).	1½+1½
	OR: New substance (1); reversibility (1); energy changes (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Definitions (2). (ii) Examples (2). (iii) New substance, gas, colour change, heat/light (1).	2+2+1
	OR: (i) Physical: boiling, melting; chemical: curd, rusting (2). (ii) Melting and boiling (1). (iii) Air plus moisture (1). (iv) Oil and dry storage (1).	2+1+1+1
20	(i) Reaction with oxygen with heat and light (1). (ii) Fuel, oxygen, ignition temperature (2). (iii) Remove fuel, cut off oxygen, cool (2).	1+2+2
	OR: (i) Oxygen is used (1). (ii) Oxygen (1). (iii) Chemical (1). (iv) Smothering fire with a blanket (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** chemical change	new product
2	**(a)** Dissolving sugar in water	recoverable
3	**(c)** ignition temperature	kindling point
4	**(b)** chemical change	new substance forms
5	**(a)** Burning of coal	cannot return
6	**(c)** bubbles of gas	carbon dioxide formed
7	**(c)** corrosion (chemical change)	new compound forms
8	**(b)** physical change	state change
9	**(d)** a physical change	state change
10	**(d)** cover it to stop air	water spreads oil fire

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A new substance, iron oxide, forms (2).	2
12	Melting of wax; dissolving sugar (2).	2
13	Cooking rice; milk turning into curd (2).	2
14	Formation of crystals from a solution (1); physical (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Melting wax is physical (1.5); burning wick/vapour is chemical (1.5).	1½+1½
16	Remove fuel (1); cut off oxygen (1); lower temperature below ignition point (1).	1+1+1
17	Reversible: melting, freezing, stretching (1.5); irreversible: burning, rusting, cooking (1.5).	1½+1½
	OR: Weathering, rusting, ripening (1.5); small reaction rates (1.5).	1½+1½
18	Food becomes unfit (1.5); refrigerate, pickle, seal (1.5).	1½+1½
	OR: Burning needs oxygen (1.5); without oxygen flame goes out (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Iron + oxygen + water (1). (ii) Test tubes experiment (2). (iii) Paint, galvanise, oil (2).	1+2+2
	OR: (i) Five changes with classification (3). (ii) Desirable or undesirable (2).	3+2
20	(i) Fizzing (1). (ii) Carbon dioxide (1). (iii) Turns milky (1). (iv) Yes (2).	1+1+1+2
	OR: (i) Heat provides energy for reaction (2). (ii) Burning, cooking (2). (iii) Lowest temperature to catch fire (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.